

The Engine Room

Part 1: Texts and Exercises

A Marine Engineer

Exercise 1

valve	a device that permits the flow of liquid or gas
bolt	a grooved metal rod with a head used with a nut to hold parts together
screw	a sharply grooved metal rod that can be driven in to hold an object together
filter	a device for removing dirt or other particles from air or liquid
bearing	a machine part inside or around which another part revolves
machine	a structure with many elements for doing some work
tube sleeve	a tube part fitting around another part
nut	a small metal piece with a hole for fastening a bolt

Exercise 2

practical	training
sufficient	knowledge
machine	repairs
rusty	bolts
successful	candidate
qualified	engineer
competency	certificate
working	period

The Engine Room

Part 1: Texts and Exercises

The Engine Department

Exercise 1

Engineers take care of the ship running
Junior officers assist the chief engineer
Officers have ratings working under their supervision
The marine engineer is a seaman at heart
The job demands physical work
He has to keep documents on the machinery working
He is responsible for all the propulsion machinery
His duties include cleaning of the engine room

Exercise 2

1. GREASER
2. VESSEL
3. SPARE
4. REPAIR
5. DOCUMENT
6. PUMPMAN
7. SYSTEM
8. OFFICER
9. BOILER
10. MAINTENANCE
11. LOG
12. CHIEF
13. AUTHORITY
14. PETTY

Exercise 3

The captain of pirates

The Engine Room

Part 1: Texts and Exercises

Watch Keeping

Exercise 1

- 1) twenty-four hours to four hours
- 2) four hours to eight hours
- 3) eight hours to twelve hours
- 4) twelve hours to sixteen hours
- 5) sixteen hours to eighteen hours
- 6) eighteen hours to twenty hours
- 7) twenty hours to twenty-four hours

Exercise 2

- 1) seeing
- 2) tasting
- 3) touching
- 4) hearing
- 5) smelling

The Engine Room

Part 1: Texts and Exercises

Internal Combustion Engines

Exercise 1

- 1) temperature
- 2) petrol
- 3) physics
- 4) petrol
- 5) expanding
- 6) burning

Exercise 2

ENGINE PARTS

piston
piston top
crankshaft
rod end
connecting rod
cylinder

ENGINE OPERATION

combustion
energy
propulsion
compression
power
drive

The Engine Room

Part 1: Texts and Exercises

Four-stroke Cycle

Exercise 1

1. heat up
2. force
3. powers
4. draws
5. convert
6. draws

Exercise 2

convert	change
tiny	very small
squeeze	compress
increase	rise
turn around	rotate
atomised	vaporised
complete	finish
drawn	pushed into
drop	fall
charge	load

The Engine Room

Part 1: Texts and Exercises

Two-stroke Cycle

Exercise 1

1. TRUE
2. FALSE
3. FALSE
4. FALSE
5. TRUE
6. TRUE

Exercise 2

1. efficient
2. consumption
3. shorter
4. loads
5. operation
6. unsuitable
7. range
8. powerful
9. revolution

The Engine Room

Part 1: Texts and Exercises

Four-stroke Diesel Engine

Exercise 1

IV stands for **injection valve**

CC stands for **combustion chamber**

EV stands for **exhaust valve**

SP stands for **spark plug**

PR stands for **piston ring**

P stands for **piston**

CR stands for **connecting rod**

CS stands for **crankshaft**

Exercise 2

The cylinder head **houses** fuel injectors.

The crankpin **forms** a part of a crankshaft.

The cylinder head **covers** the cylinder liner from the top.

A spark plug **ignites** the air-fuel mixture.

The piston **moves** up and down inside the cylinder.

A pushrod **transmits** the action of a cam to a valve.

A 4-stroke **consumes** less lubricating oil.

The engine block **supports** cylinder liners.

The piston pin **joins** the connecting rod to the piston.

Fuel **enters** through fuel injectors at high velocity.

Exercise 3

1. CYCLE
2. STROKE
3. RPM
4. TEMPERATURE
5. PORT
6. HEAT
7. THRUST
8. AXIS
9. PISTON
10. ROD
11. INTAKE

The Engine Room

Part 1: Texts and Exercises

Two-stroke Diesel Engine

Exercise 1

ENGINE CONSTRUCTION

secured to
mounted on
supported by
joined to
located in
connected to

ENGINE OPERATION

reduced by
compressed by
scavenged by
used to
injected into
drawn into
expelled from

Exercise 2

Picture A.

The fresh air is drawn **<into>** the crankcase.

The piston is forced **<up>** the cylinder liner.

The exhaust port is closed **<by>** the piston.

The mixture is ignited **<by>** the spark plug.

Picture B.

The piston is forced **<down>** the crankcase.

The exhaust port is opened **<by>** the piston.

The exhaust gases are expelled **<from>** the cylinder.

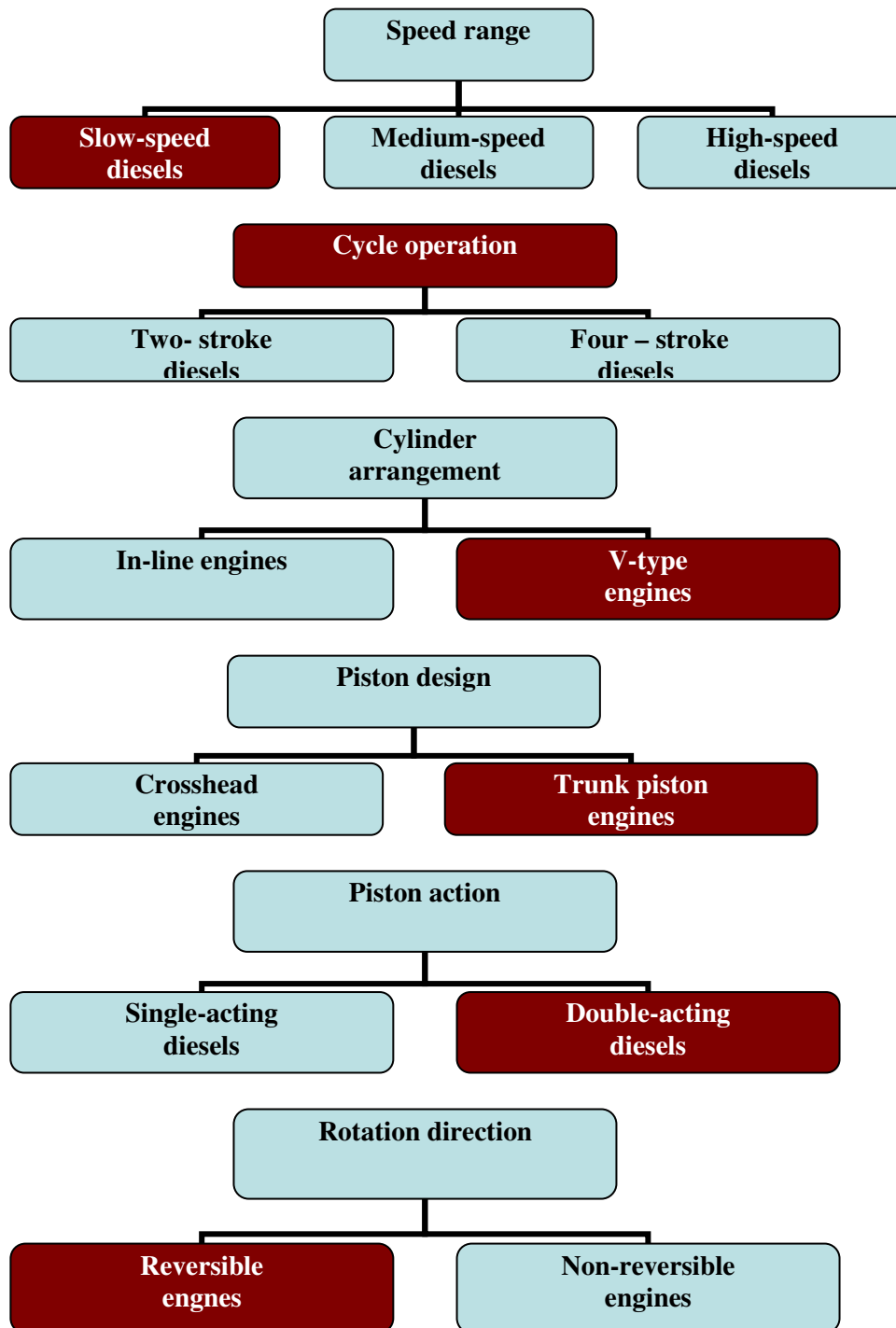
The combustion space is scavenged **<by>** the incoming air.

The Engine Room

Part 1: Texts and Exercises

Diesel Engine Types

Exercise 1



Exercise 2

1. is ignited
2. are classified
3. are used
4. is attached
5. are designed
6. is achieved
7. are coupled
8. are made

Exercise 3

What **do engines operate with**?

What mechanisms **do indirect drive engines involve**?

Where **does the piston travel**?

What **did Rudolf Diesel invent**?

Who **installs two-stroke engines on modern vessels**?

What **did Rudolf Diesel invent**?

What **do engines operate with**?

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Part 1: Texts and Exercises

Diesel Engine Parameters

Exercise 1

Bore	mm
Stroke	cm
Stroke/bore	
Number of Cylinders	
Speed	rpm
Mean Piston Speed	m/s
Mean Effective Pressure	bar
Fuel Consumption	t/day
Power/cylinder	kW

Exercise 2

Operation cycle	four-stroke
Cylinder arrangement	both
Speed range	medium-speed
Fuel kind	both
Piston design	crosshead type
Rotation direction	clockwise

Exercise 3

stroke length	350 mm
MEP	20 bars
bore	320 mm
piston speed	8.4 – 8.75 m/s
power range	1400 – 6500 kW
cylinder output	300 – 350 kW

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Part 1: Texts and Exercises

Dual Fuel Systems

Exercise 1

1. depends on
2. run
3. ensure
4. run
5. reduces
6. supplies

Exercise 2

1. camshaft-driven pump
2. spring-loaded valve
3. valve-supplied gas
4. hand-operated valve
5. health-related emissions
6. coating-equipped plunger
7. valve-actuated gas admission
8. WECS-controlled fuel amount

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Part 1: Texts and Exercises

Jerk Pump Injection

Exercise 1

high-pressure pump	pressurises the fuel
fuel injection pump	sends fuel into injection nozzle
feed pump	sucks fuel from the tank
fuel filter	removes impurities from the fuel
injection nozzle	injects fuel into the cylinder

Exercise 2

- How does the pressure drop
- How does the spring close the valve
- How is the needle valve held against its seat
- How does the fuel pressure above the plunger build up
- How does the fuel flow into the space
- How can the injection be adjusted

The Engine Room

Part 1: Texts and Exercises

Operating Procedures

Exercise 1

install	uninstall
close	open
loosen	tighten
raise	lower
mount	dismount
increase	decrease
switch on	switch off
screw	unscrew
connect	disconnect

Exercise 2

engine fails to turn on starting air

1. **start**
2. **open**
3. **lubricate**

engine turns on fuel but runs unevenly

1. **start**
2. **clean**
3. **increase**

exhaust temperature rises

1. **replace**
2. **check**

engine rpm decrease

1. **raise**
2. **overhaul**

smoky exhaust

1. **check**
2. **activate**

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Part 1: Texts and Exercises

Fuel Oil Supply System

Exercise 1

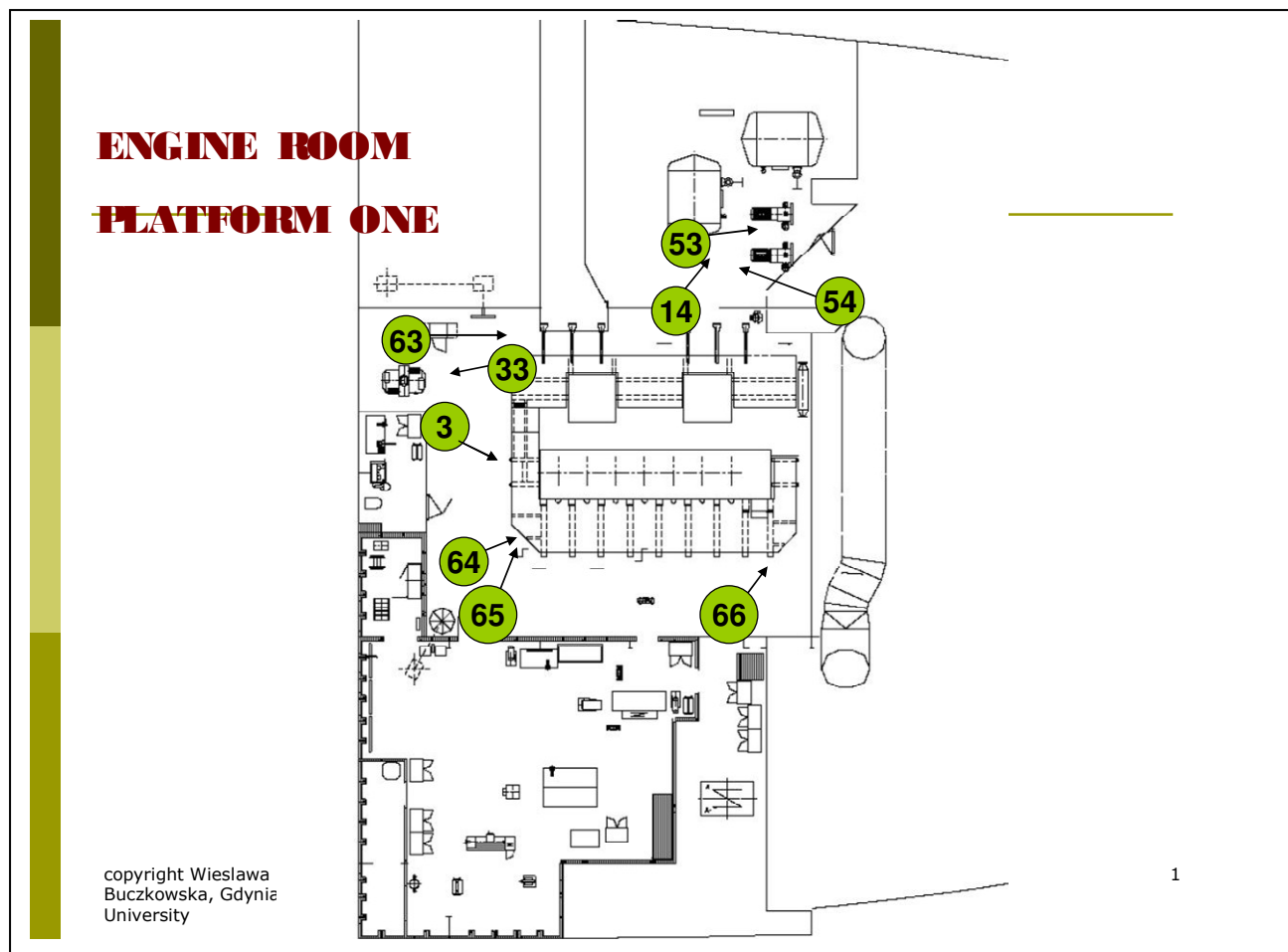
1. FALSE (* TRUE mistakenly accepted as correct answer)
2. FALSE
3. TRUE (*FALSE mistakenly accepted as correct answer)
4. TRUE (*FALSE mistakenly accepted as correct answer)
5. FALSE (*TRUE mistakenly accepted as correct answer)

*** Note!** There is an error in the MarEng Tool regarding the exercise concerned.

The Engine Room

Part 2: The Engine Room Layout

Platform One



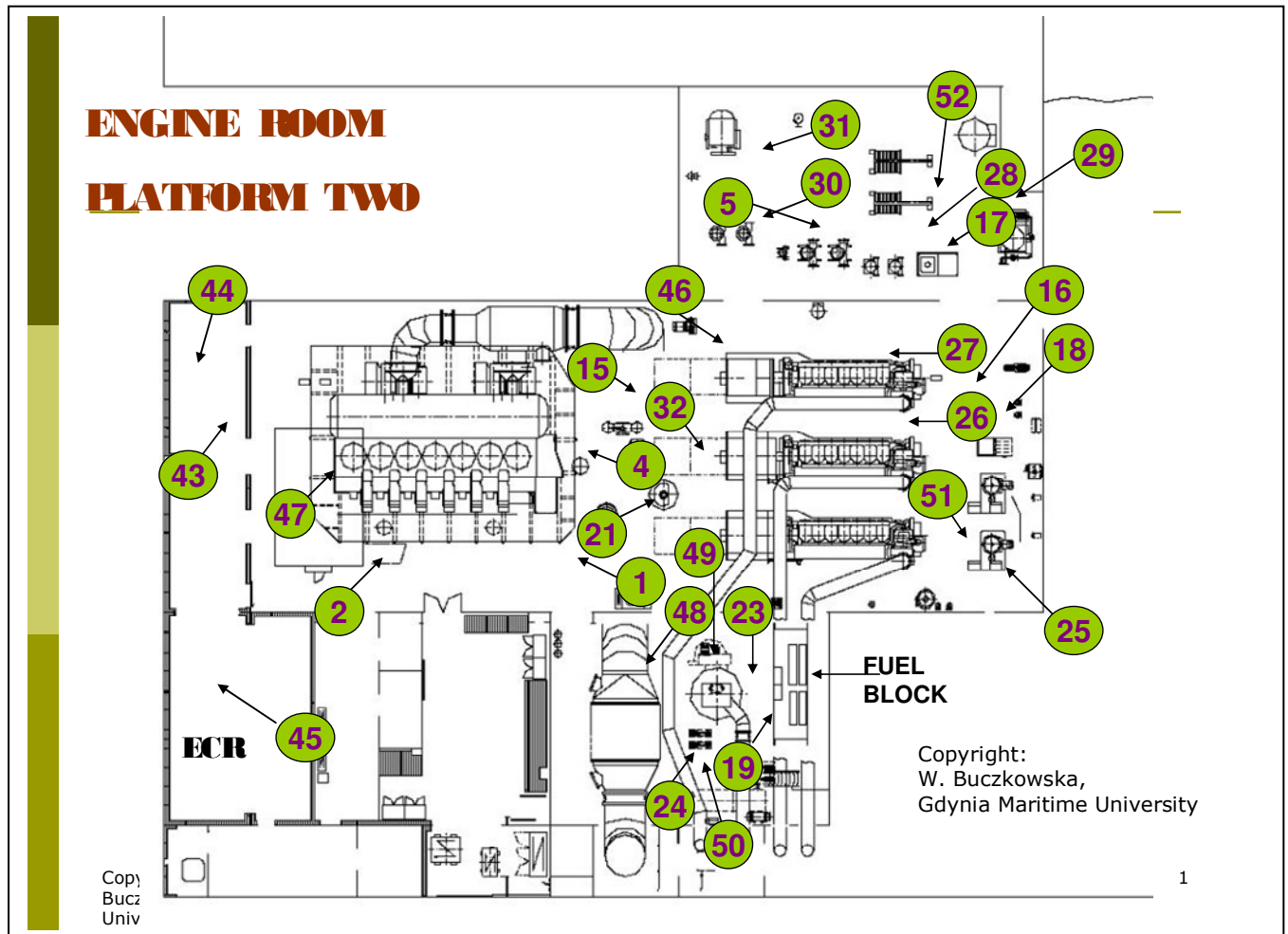
- 54. propulsion, pressurized
- 53. doubled
- 14. maintenance
- 33. ---
- 63. ---
- 3. access, clear
- 64. Main Engine
- 65. rotate
- 66. failure

Note! The picture on the MarEng CD is turned 90 degrees counterclockwise, whereas the one in the Answer Key follows the proper "bow-stern" direction.

The Engine Room

Part 2: The Engine Room Layout

Platform Two

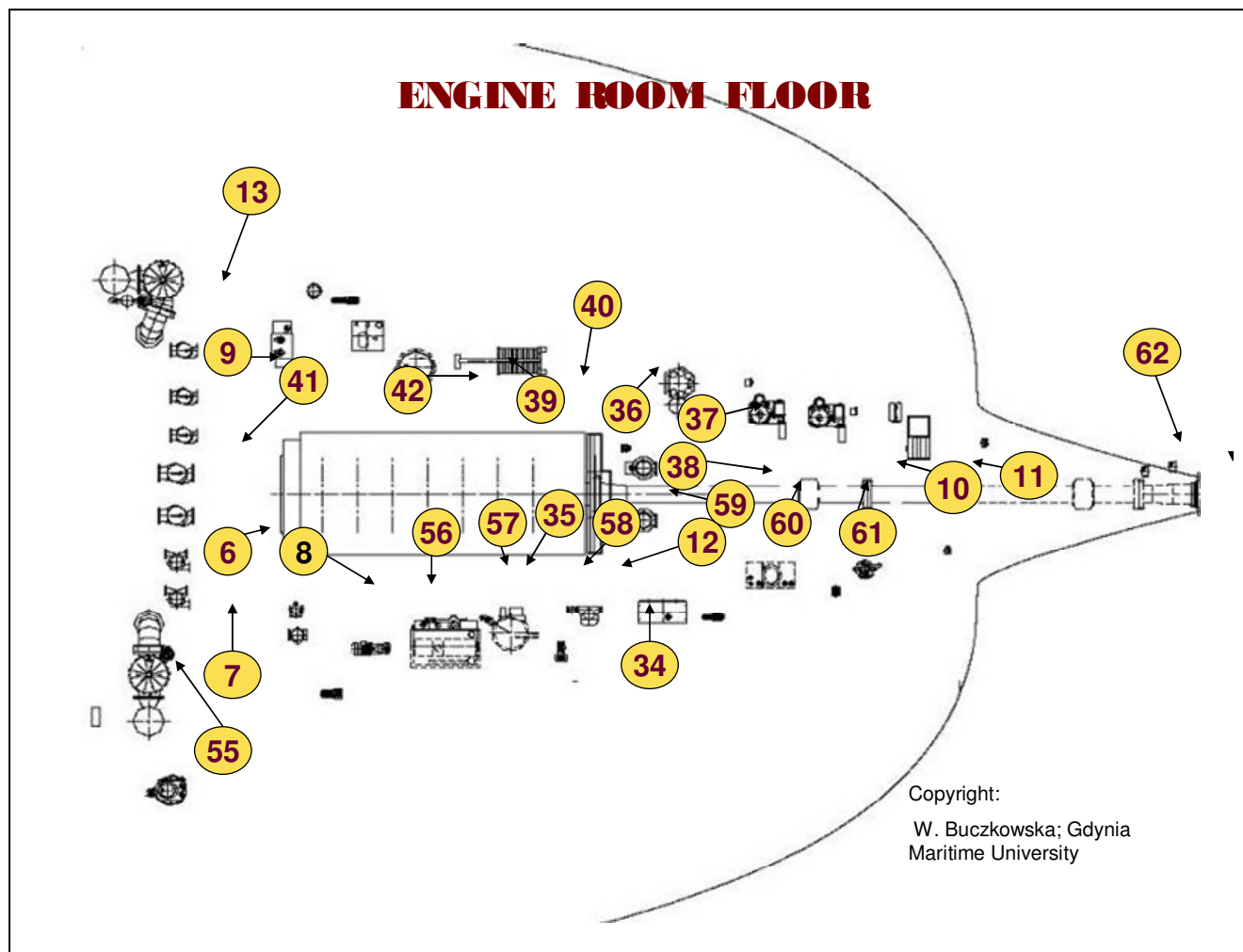


- | | | | |
|-----|----------------------------|-----|---------------------------|
| 44. | breakdowns | 15. | --- |
| 43. | speed, handle | 46. | Auxiliary Gen Sets |
| 47. | cover, four | 5. | --- |
| 2. | Cylinder Covers | 30. | Fresh Water Cooling Pumps |
| 45. | electricity | 31. | applications |
| 50. | discharge | 52. | Fresh Water Coolers |
| 24. | ---- | 28. | designed |
| 19. | Fuel Block | 29. | non-stop |
| 23. | houses | 17. | circulates |
| 48. | process | 27. | Auxiliary Gen Sets |
| 49. | burning | 16. | mounted |
| 1. | Main Engine Cylinder Heads | 18. | Fuel Oil Separator |
| 21. | passages | 26. | Auxiliary Gen Sets |
| 4. | collected | 51. | processes, cause |
| 32. | ---- | 25. | continuous, produce |

The Engine Room

Part 2: The Engine Room Layout

Engine Room Floor



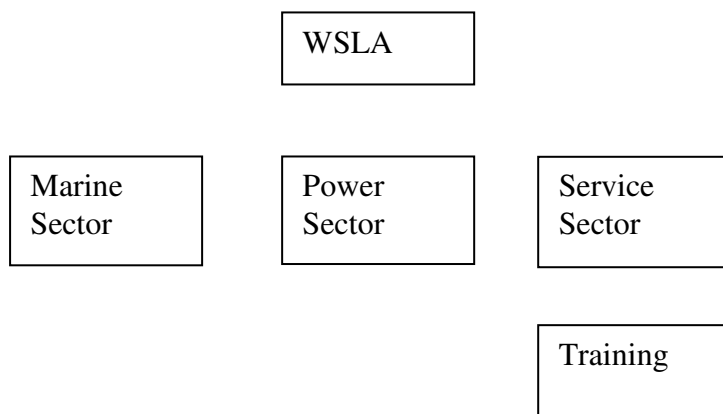
- | | | | |
|-----|------------------------|-----|---|
| 62. | prevent | 57. | contains |
| 11. | consists of, measuring | 56. | outboard |
| 10. | crankshaft | 8. | sewage treatment plant,
oily water separator,
piston bilge pump |
| 61. | circular | 6. | present |
| 60. | rotating | 7. | mounted, flows |
| 34. | remove | 55. | located |
| 12. | force, emergency | 39. | main engine lub oil cooler |
| 59. | minimise, raise | 42. | lub oil cooler |
| 38. | manufactured | 41. | disastrous |
| 37. | rotates | 9. | avoid, corrugated |
| 36. | changeover | 13. | ballast pumps, sea chest |
| 40. | separate, reduces | | |
| 58. | decreasing | | |
| 35. | discharging, -stage | | |

The Engine Room

Part 3: Texts, Videos and Exercises

Interview with Peter Lund from Wärtsilä Land and Sea Academy

Exercise 1



Exercise 2

We have full **scope** training.

It started off with the training **switch** in mainly providing product training internally.

Because we have this global **infrastructure**, we have a **global** advantage over individual training institutions, which are **resident** in one country.

Exercise 3

According to Peter Lund, the **<regulations>**, people management and **<allocation/allocating>** of resources are the most important tasks of a Chief Engineer.

Having technical **<knowledge>** is good, but sometimes very difficult because machinery changes so much.

It is hard to, on the **<spur>** of the moment, find managing skills and knowledge to **<satisfy>** regulatory bodies like **<survey>** authorities.

As far as growing the mind, there's nothing like travel for **<broadening>** the mind.

The Engine Room

Part 3: Texts, Videos and Exercises

Fuel Bacteria Results in Grounding

Exercise 1

1. **When the Loch Striven was to begin its voyage, she was manned with**
B only a part of her normal crew on board.
2. **During the first day of the voyage**
C no problems were experienced.
3. **When the aft main engine stopped, the crew was**
C changing the course of the vessel.
4. **Why were the fuel filters blocked at this particular time?**
B The movement of the vessel shook loose the deposits.
5. **Why did the master of the vessel not take action to avoid the engine problems?**
A The problem had not been reported to the master of the vessel.
6. **The fuel filters on board the Loch Striven were**
A easy to change.
7. **When handing over a vessel to a new crew,**
B problems that might recur must be reported.
8. **Fuel contaminated by bacteria can be avoided by**
A draining the fuel tanks.

Exercise 2

break	away
line	up
change	over
close	up
make	sure
allow	for
tie	up

The Engine Room

Part 3: Texts, Videos and Exercises

Engine-Room Fire alongside Oil Berth

Exercise 1

1. A rocker arm punctured the engine entablature and caused the fire. **(F)**
2. The vessel's fire party extinguished the fire. **(F)**
3. The emergency fire pump did not get water because the draught forward was so small **(T)**
4. For various reasons, the auxiliary engine ran for an extended period of time. **(T)**
5. The engine room was being painted and the paint contributed to the intensity of the fire. **(F)**
6. One of the crew members stayed in his cabin because he did not hear the alarm. **(F)**

Exercise 2

1. Pressure-test the CO₂ manifold. The CO₂ manifold **<has been pressure-tested>**.
2. Conduct periodic checks. Periodic checks **<have been conducted>**.
3. Confirm the effectiveness of the remote control wire for the CO₂ release. The effectiveness of the remote control wire for the CO₂ release **<has been confirmed>**.
4. Give an assurance that the emergency fire pump will always be available. An assurance that the emergency fire pump will always be available **<has been given>**.
5. Issue standing orders for crew to respond promptly to any emergency alarm. Standing orders for crew to respond promptly to any emergency alarm **<have been issued>**.

The Engine Room

Part 3: Texts, Videos and Exercises

Engine Room Tools, exercises

Exercise 1

SCREWCLAMP
O-RING
STEELBRUSH
CUTTING PLIERS
PUNCH PIN

Exercise 2

SPANNER
TORQUE WRENCH
ROUND FILE
RING SPANNER
OIL STICK

The Engine Room

Part 3: Texts, Videos and Exercises

A Maintenance Procedure

Exercise 1

- A. **Prestress the spring slightly by tightening the adjusting screw.**
- B. Open the manometer valve.
- C. Adjust the opening pressure to 50-100 bar.

- A. The nozzle has to be repaired or replaced if any drops occur on the nozzle tip.
- B. Increase the pressure in the test pump.
- C. **Connect the injection valve to the test pump and pump to expel air.**

- A. **Shut the manometer valve and pump rapidly to blow dirt out of the nozzle orifices.**
- B. Connect the injection valve to the pump.
- C. Tighten the cap nut.

- A. **Adjust the opening pressure to 50-100 bar.**
- B. Keep pressure constant for 10 seconds.
- C. Tighten the cap nut.

- A. Fit the injection pipe in place.
- B. **Put a dry paper under the nozzle.**
- C. Pump slowly and watch the manometer to note the nozzle opening pressure.

- A. Stop pumping and measure the pressure drop time
- B. **Give the pump a quick blow.**
- C. Check that no fuel drops occur on the nozzle tip.

- A. Clean and check the parts carefully.
- B. Check the high pressure sealing faces of the nozzle holder.
- C. **Check the uniformity of spray marks on the paper.**